

Muscle Growth Is Very Strongly Correlated with Strength Gains after Lower Body Resistance Training: New Insight from Within-Participant Associations

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ABSTRACT

MARQUES, E. A., T. G. BALSHAW, M. P. FUNNELL, E. J. MCDERMOTT, S. MAEO, L. J. JAMES, and J. P. FOLLAND. Muscle Growth Is Very Strongly Correlated with Strength Gains after Lower Body Resistance Training: New Insight from Within-Participant Associations. *Med. Sci. Sports Exerc.*, Vol. 57, No. 12, pp. 2838–2845, 2025. **Purpose:** Quantitative mechanistic evidence for muscle size and agonist neuromuscular activation explaining the individual strength gains that occur after resistance training (RT) remains opaque and controversial, with nonsignificant or weak/moderate relationships reported. This study aimed to quantify the within-participant association of strength gains following 15 wk of RT with muscle growth and the changes in agonist neuromuscular activation. **Methods:** Knee extension muscle strength (isometric maximum voluntary torque (iMVT) and single repetition maximum (1RM)), magnetic resonance imaging–derived quadriceps muscle volume, and normalized (to Mmax) quadriceps surface electromyography amplitude were measured in 39 previously untrained healthy young males, before and after 15 wk of lower body RT (3× per week). Data were analyzed using repeated-measures (within-participant) correlations and linear mixed models. **Results:** Very strong repeated-measures correlations were found between muscle growth and strength gains (iMVT: $r = 0.92$, $P < 0.001$; 1RM, $r = 0.89$, $P < 0.001$). Changes in surface electromyography amplitude were moderately correlated with the changes in strength (iMVT: $r = 0.58$, $P < 0.001$; 1RM: $r = 0.56$, $P < 0.001$). Linear mixed models revealed muscle growth and changes in neuromuscular activation both had significant positive effects on strength gain, but muscle growth contributed >5-fold more than neuromuscular activation (standardized beta coefficient = 0.88–0.94 vs 0.13–0.17). **Conclusions:** Using more powerful within-participant statistical techniques, these results show that muscle growth is, in fact, very strongly, and neuromuscular activation moderately associated with strength gains after RT. These findings suggest that muscle growth is a major adaptation that is highly relevant to the individual increases in strength of young healthy males after RT. **Key Words:** HYPERTROPHY, IMAGING, MAGNETIC RESONANCE, MUSCLE VOLUME, NEUROMUSCULAR ACTIVATION, sEMG

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Resistance training (RT) is widely recommended for all age groups and levels of physical activity/function (1). It is particularly beneficial for athletic development (2), prevention and rehabilitation of musculoskeletal and sports injuries (3), and healthy aging (4), including numerous musculoskeletal conditions, such as osteoarthritis (5), osteoporosis (6), and sarcopenia (7). Therefore, understanding the physiological adaptations responsible for RT-induced strength gains is imperative to refine RT and optimize the desired outcomes. Currently, there is extensive evidence that both muscle growth/hypertrophy and neuroplasticity occur after regular RT (8,9), with these changes extensively conceptualized as the primary determinants of strength gains (10,11). However, the functional relevance of these neural and morphological adaptations

and their contribution to individual strength gains remains somewhat opaque and controversial (12–14). We hypothesized that a range of methodological issues in measuring these adaptations and analyzing their statistical association account for this lack of clarity and, if corrected, may reveal a clearer, more pronounced relationship between these putative primary mechanisms and individual strength gains.

Despite the between-participant muscle size–strength relationship having been extensively documented over decades (15), with, for example, muscle volume accounting for ~60% of the interindividual variability in isometric strength (16), the contribution of muscle growth to strength changes following RT remains controversial (12,13,17–19). In part, perhaps due to the surprisingly limited evidence and typically insignificant–modest relationships reported between the changes in muscle size and strength after RT (e.g., $r = 0.03$ – 0.53) (9,20–24). The reasons for this debate and the surprisingly modest relationship between muscle growth and strength gains after RT may include, firstly, muscle size measurements with low-resolution techniques (e.g., ultrasound/DXA) that lack the sensitivity for precisely delineating the relationship (20,22). However, two studies from our laboratory utilizing high-resolution magnetic resonance imaging (MRI), the gold standard method of determining muscle size, have also revealed relatively modest between-participant relationships between muscle strength increases and size gains after 12 wk of RT ($r = 0.46$ – 0.53) (9,24). A second possibility is the limited muscle growth reported by relatively short-duration RT studies (i.e., 8–12 wk (9,20,22–24)) when neural adaptations, including enhanced agonist neuromuscular activation (11), may predominate during this early phase of RT (24). Therefore, longer-duration RT studies that elicit a greater muscle growth response may better delineate the muscle growth–strength gain relationship. Thirdly, individual strength change data may be particularly sensitive to measurement errors at both time points, and thus, duplicate measurements both pre- and post-RT may improve the precision of the measured changes (25).

Although a wide range of neural changes have been observed after RT, the functional relevance of these changes remains highly uncertain, such that a recent review of current evidence concluded that changes in motor performance and neuroplasticity tend to be uncoupled (14), and these authors called for more rigorous approaches. Furthermore, specific to the changes in surface electromyography (sEMG) amplitude after RT, one common measure of agonist neuromuscular activation, the majority of previous studies did not report or find an association between sEMG changes and strength gains (14). However, changes in absolute sEMG amplitude measurements may have been confounded by increases in muscle size, as we recently found absolute sEMG amplitude measurements to be affected by muscle size (26). To overcome this issue, in a previous study, we assessed the relationship of normalized sEMG amplitude (to Mmax to control for peripheral changes, e.g., muscle hypertrophy) with strength gains after 12 wk of isometric RT and found a moderate correlation ($r = 0.576$) (24). However, this study included participants who completed two quite distinct training regimes that led to

differential adaptations and may have inflated the association of strength gain and normalized sEMG amplitude. Therefore, further examination of the relationship of neuromuscular activation, alone and in combination with muscle growth, with strength gains appears warranted.

Importantly, the statistical analysis could also affect our understanding of the relationship between strength gains and putative underpinning variables. Previous research has heavily relied on the use of between-participant analyses, such as simple regression or Pearson's correlation, to investigate the relationships between changes in muscle size or activation with changes in strength. These methods assess interindividual associations but may lack sensitivity in capturing within-participant changes over time. For example, the most common approach has been to consider the percentage change (i.e., pre-training to post-training) in strength and size/activation over a training period, using one data point per subject (e.g., (9,20–24,27–30)), which may obscure the association due to the aggregation of the two dependent time points (22,31,32). In other words, relatively fixed individual factors (e.g., moment arm, contractile specific tension) may be substantial, such that individual pre and post values are dependent and better considered together. Thus, when two or more measurement points are obtained from the same individual, the within-participant (or repeated-measures) correlation is the preferred statistical method for analyzing the common intraindividual association (33). Because repeated-measures correlation accounts for the non-independence of each paired data, it tends to yield much greater power than data that are averaged or derived from the changes between time points (31).

Hence, this study aimed to assess the intraindividual relationship between the changes in muscle volume and neuromuscular activation with the changes in strength (isometric and 1-repetition maximum lifting strength (1RM)) in male adults after a relatively prolonged 15 wk (45 sessions) of RT. To maximize the precision of the measurements, muscle volume was measured with MRI, neuromuscular activation was assessed with sEMG normalized to Mmax, and strength and normalized sEMG were assessed in duplicate at both pre- and post-RT to minimize measurement error. We hypothesized that the intraindividual correlation coefficients would have a larger magnitude than using interindividual correlation and that neuromuscular activation changes would be the biggest predictor of strength gains after RT rather than changes in muscle size (24).

METHODS

Data Source

We used data from a 15-wk single-center randomized, double-blind placebo–control trial from a previously published study (34). Thirty-nine men with no prior lower-body injuries, no recent history (>18 months) of RT, a low-to-moderate level of recreational physical activity, but not involved in regular physical training, aged 18 to 40 yr (mean $\pm$ SD age, 26 ± 4 yr; height, 1.79 ± 0.08 m; body mass, 74.6 ± 11.7 kg; body mass index, 23.1 ± 2.8 kg·m⁻²), and with adequate protein and energy

intake, completed the intervention. The study methods and the primary outcomes for the two groups—collagen peptide ($n = 19$) and placebo ($n = 20$) supplementation—have been described in detail (34). Briefly, participants consumed either collagen peptide or placebo supplements once daily during the 15-wk RT intervention period (105 daily doses) and were instructed to maintain their habitual physical activity and usual diet throughout the study. All participants volunteered for this study and provided written informed consent prior to their participation. The study was approved by the Loughborough University Ethics Review Sub-Committee (no. R19-P030). As this study was a secondary analysis, no power analysis was performed.

RT Protocol

In brief, exercise sessions were performed on nonconsecutive days, $3 \text{ d} \cdot \text{wk}^{-1}$, and each session lasted approximately 45 min for a period of 15 wk. All participants completed 45 training sessions of the same standardized RT program that exclusively focused on the lower body, targeting primarily the quadriceps and secondarily the hamstrings and gluteus maximus of both legs. Each session involved completing 2–4 sets of each of three isoinertial (constant load) exercises in the following order: unilateral knee extension (KE) (TechnoGym, Selection Leg Extension, Bracknell, UK), bilateral knee flexion (KF) (Life Fitness, Seated Leg Curl SL40, Cambridgeshire, UK), and bilateral leg press (Watson Gym Equipment, 45° Leg Press, Somerset, UK). A 2-min recovery period separated sets with the same leg/legs, and a bilateral warm-up set was also completed before the KE and KF exercises. The number of sets progressively increased during the first 6 wk of the intervention period, with four sets being completed for all three exercises by week 7. The load and number of repetitions per set were varied according to an undulating periodized program between ~ 12 repetition maximum (RM) to ~ 6 RM (see Supplemental Table 1, Supplemental Digital Content, <http://links.lww.com/MSS/D282>).

Measurements

A familiarization session, involving all the contractions/tasks to be assessed, was completed ~ 7 d before the first measurement session. Identical measurement sessions were completed twice (3–5 d apart) before and twice after the RT (2–3 and 4–6 d after the last training session). All the measurements were unilateral and of the dominant leg. Each session involved strength (both isometric and 1RM lifting strength, in this order), as well as simultaneous neuromuscular activation (sEMG) assessments. Muscle volume was assessed with MRI before (5–7 d prior to the first training session) and after RT (3–5 d after the final training session), and the scan time was standardized within participants. A concise description of the measures relevant to this manuscript is provided here. For a more thorough description of the methodology, see Ref. (34).

KE strength. Both strength tests were preceded by an incremental warm-up of contractions (~ 5 s contractions at 50% ($\times 3$), 75% ($\times 3$), and 90% ($\times 1$) of perceived maximum) or lifts (50% previous 1RM $\times 8$ repetitions; 75% previous 1RM $\times 5$

repetitions; 85% previous 1RM $\times 2$ repetitions; with 1 min between sets). Isometric KE strength was assessed while participants were seated on a rigid custom-made isometric dynamometer with knee and hip joint angles of 104° and 126° (180° = full extension). Participants completed 3–4 maximum voluntary contractions of the dominant leg and were instructed to “push as hard as possible” (extension) for 3–5 s, with ≥ 30 -s rest between each effort. Their greatest instantaneous torque was defined as KE isometric maximum voluntary torque (iMVT). KE 1RM was assessed using the same machine as during training, with the participant’s hips firmly strapped down and the handles held on either side of the seat. The first single lift was performed at the previous 1RM load (from the preceding familiarization or measurement session); if successful, the load was increased by 2.5 to 5.0 kg for the next lift. 1RM was established typically within three to four attempts (single lifts, separated by 2-min recovery) to the nearest 2.5 kg, with additional attempts performed if necessary.

Agonist neuromuscular activation. sEMG was recorded from the superficial quadriceps (vastus lateralis (VL), vastus medialis (VM), and rectus femoris (RF)) during both strength tests and femoral nerve stimulation, using a wireless EMG system (Trigno; Delsys Inc., Boston, MA). Two sensors were placed over each superficial quadriceps muscle at set percentages of thigh length (the distance between the lateral knee joint space and greater trochanter) above the superior border of the patella as follows: RF 65% and 55%; VL 60% and 55%; VM 35% and 30%. Sensors were placed parallel to the presumed orientation of the underlying fibers. EMG signals were amplified at the source ($\times 300$; 20- to 450-Hz bandwidth) before further amplification (overall effective gain, $\times 909$). EMG signals were sampled at 2000 Hz via the same A/D converter and computer software as the force signal, enabling data synchronization.

Root-mean-square (RMS) EMG amplitude of each quadriceps sensor was measured during a 500-ms epoch surrounding KE iMVT (250 ms on either side). Mean RMS EMG (200 ms moving window) during the heaviest successful 1RM was measured between the initial start position and most extended lever arm position (assessed by recording a potentiometer signal of the lever arm position). RMS EMG values from each quadriceps sensor at iMVT and during the 1RM were normalized to the corresponding peak-to-peak maximal M-wave response elicited during supramaximal femoral nerve stimulation (for details, see (34)) before averaging across sensors to produce an overall normalized agonist EMG measurement during each task, which was used in the analysis.

Muscle volume. T1-weighted axial and coronal plane MRI scans (3.0 T Discovery MR750w; GE Healthcare, Chicago, IL) of the dominant thigh (between the anterior superior iliac spine to the lateral tibial condyle) were used to assess the volume of the quadriceps muscles. Using a receiver eight-channel whole-body coil, two overlapping blocks of both axial images (time of repetition/time to echo, 600/8.35 ms; image matrix, 512×512 ; field of view, $260 \text{ mm} \times 260 \text{ mm}$; pixel size, $0.508 \text{ mm} \times 0.508 \text{ mm}$; slice thickness, 5 mm; interslice gap, 0 mm) and coronal images

(time of repetition/time to echo, 600/8.53 ms; image matrix, 256 × 256; field of view, 450 mm × 450 mm; pixel size, 1.76 mm × 1.76 mm; slice thickness, 5 mm; interslice gap, 0 mm) were acquired, allowing an objective alignment during analysis (Horos, version 3.3.6, www.thehorosproject.org).

The muscle volume of the four constituent muscles of the quadriceps (RF, VL, VM, and vastus intermedius (VI)) was analyzed. One trained investigator manually segmented every third image (i.e., every 15 mm) along each muscle from the first distal image in which that muscle appeared until the muscle was no longer visible (mean 27–29 slices/images per individual muscle) using Horos software (version 3.3.6, www.thehorosproject.org). The volume of each muscle was calculated as the area under the anatomical cross-sectional area–muscle length curve following cubic spline interpolation with 1000 points (GraphPad Prism 8; GraphPad Software). Constituent muscle volumes were summed to determine the volume of the whole quadriceps. Within-participant between-day reliability of repeated scans and analyses to assess quadriceps volume in our laboratory was 0.9% (34).

Statistical Analysis

For duplicate measurements at each time point, average values of the two sessions were used in the analysis. Descriptive outcomes are presented as mean ± standard deviation, and the mean differences between pre- and post-values were tested using paired *t*-tests. To test our hypotheses, we first examined if there was a significant group effect (collagen peptide vs placebo supplements) on the association of strength gains with either muscle growth or the changes in agonist neuromuscular activation. To do this, we used linear mixed-effects models (estimated using restricted maximum likelihood (REML)), adjusting for group and with each individual as a random effect (31,35). As no significant group effects were observed on these associations, data from both groups were subsequently pooled and analyzed together. Repeated-measures correlations (i.e., within-participant) were performed using the R package–labeled “rmcorr” (31), as the final models did not include any covariate adjustments. Rmcorr provides the best linear fit for each participant using parallel regression lines. Like a Pearson correlation coefficient, the rmcorr coefficient is bounded by –1 to 1, representing the strength of the linear association between two variables (31). The variance was assessed with 95% confidence intervals (CIs) computed using the optional parameter proposed by the rmcorr package. The following criteria were adopted to interpret the magnitude of the correlation (*r*) for paired measures: ≤0.40, weak; 0.40–0.60, moderate; 0.60–0.80, strong; and 0.80–1.0, very strong (36).

For comparative purposes (i.e., with the repeated-measures correlation and previous literature), individual percentage changes for each variable were calculated as the change divided by the pre-intervention value multiplied by 100 and used to perform between-participant bivariate Pearson’s product–moment correlations. Finally, we used linear mixed models

(estimated using REML) to predict strength changes (KE 1RM and iMVT) from quadriceps volume and neuromuscular activation changes after RT. The models included each individual as a random effect, and the beta coefficients (β), standardized beta coefficients (Std. β), 95% CIs, and *P* values were computed using a Wald *t*-distribution approximation. Multicollinearity among the predictor variables was assessed using the variance inflation factor, and all values were found to be low (~1.1), indicating no significant multicollinearity issues in the model. *P* value <0.05 was considered statistically significant. All analyses and figures were performed using R version 4.4.0 (packages lme4, rmcrr, easystats, and ggplot2).

RESULTS

Neuromuscular characteristics at baseline and after 15 wk of RT are presented in Table 1. All outcomes significantly increased (*P* < 0.001) after the intervention. Within-participant relationships between the changes in quadriceps muscle volume and the changes in both measures of KE strength were very strong (iMVT: *r* = 0.92, 0.86–0.96, *P* < 0.001, Fig. 1A; 1RM: *r* = 0.89, 0.81–0.94, *P* < 0.001, Fig. 1B). Moderate within-participant relationships were also found for the changes in quadriceps neuromuscular activation and the changes in both iMVT (*r* = 0.58, 0.32–0.76, *P* < 0.001, Fig. 2A) and 1 RM (*r* = 0.56, 0.29–0.75, *P* < 0.001, Fig. 2B).

Between-participant correlation analysis (i.e., for comparison with previous studies) showed that the percentage changes in quadriceps volume were moderately correlated with the percentage changes in iMVT (*r* = 0.60, *P* < 0.001) and weakly correlated with the percentage changes in KE 1RM (*r* = 0.35, *P* = 0.029). The percentage change in neuromuscular activation was moderately correlated with the percentage changes in KE iMVT (*r* = 0.45, *P* = 0.004) and with KE 1RM (*r* = 0.42, *P* = 0.009).

Linear mixed models exploring within-participant changes in KE iMVT revealed significant and positive effects for both quadriceps volume change (β = 0.12; 95% CI = 0.11–0.14, *t* (73) = 13.70, *P* < 0.001; Std. β = 0.94; 95% CI = 0.81–1.08) and quadriceps neuromuscular activation change (β = 3.17, 95% CI = 1.01–5.33, *t*(73) = 2.93, *P* = 0.005; Std. β = 0.17, 95% CI = 0.05–0.28). Similarly, the linear mixed model results for changes in KE 1RM showed significant and positive effects for both quadriceps volume change (β = 0.03, 95% CI

TABLE 1. Maximum KE strength and quadriceps volume and EMG outcomes before and after 15 wk of RT (mean ± SD, *n* = 39).

	Pre-Training	Post-Training	Change (%)
Maximum strength			
KE iMVT (N·m)	228.4 ± 43.6	276.5 ± 50.5	21.6 ± 9.2
KE 1RM (kg)	52.2 ± 11.4	66.7 ± 13.7	28.6 ± 12.9
Muscle size			
Quadriceps volume (cm ³)	1980.2 ± 381.3	2217.7 ± 383.2	12.7 ± 7.1
Agonist neuromuscular activation (%M _{MAX})			
Quadriceps EMG _{iMVT}	8.5 ± 2.7	9.9 ± 2.7	22.9 ± 35.7
Quadriceps EMG _{1RM}	7.9 ± 2.4	9.1 ± 2.0	22.8 ± 35.9

All listed variables showed significant increases (*P* < 0.001), as determined by paired *t*-tests comparing pre- and post-training values.

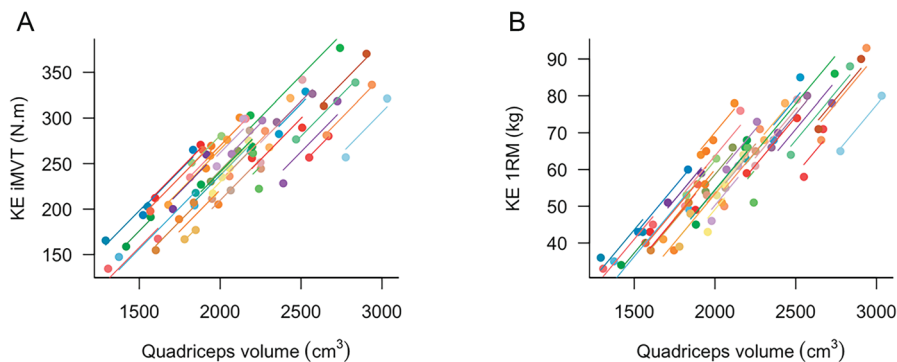


FIGURE 1—Within-participant relationships ($n = 39$) between the changes in quadriceps volume and the changes in (A) KE iMVT ($r = 0.92$) and (B) KE 1RM ($r = 0.89$) over 15 wk of RT. All figures display observations from the same participant using the same color, with corresponding lines to show the repeated-measures correlation fit for each participant.

$= 0.03$ – 0.04 , $t(71) = 16.07$, $P < 0.001$; Std. $\beta = 0.88$, 95% CI = 0.77 – 0.99) and quadriceps neuromuscular activation change ($\beta = 0.82$, 95% CI = 0.14 – 1.50 , $t(71) = 2.42$, $P = 0.018$; Std. $\beta = 0.13$, 95% CI = 0.02 – 0.24).

DISCUSSION

To our knowledge, this study is the first to use repeated-measures correlation to investigate whether RT-driven changes in muscle size and neuromuscular activation contribute to strength gains during prolonged RT. In contrast to previous findings, the present results of repeated-measures correlations demonstrated that strength gains and muscle growth are, in fact, very strongly related ($r = 0.89$ – 0.92) and substantially more strongly related than previous findings. Contrary to our hypothesis, strength gains and muscle growth were also more strongly related than the changes in neuromuscular activation and strength gains ($r = 0.56$ – 0.58). Similarly, although both the changes in muscle size and activation contributed to strength gains within the linear mixed models, muscle growth was the predominant contributing predictor, being >5 times as strong a predictor of strength gains in both tasks as changes in activation (Std. β 0.88 – 0.94 vs 0.13 – 0.17). Interestingly, between-participant correlations revealed markedly weaker relationships of strength gains with the changes in muscle size

($r = 0.35$ – 0.60) and activation (0.42 – 0.45) than for repeated-measures correlation, and thus, the statistical approach appears critical in correctly determining the nature of the relationship of strength gains with changes in muscle size and activation. Although causation remains unproven, the findings add emphasis to the importance of gains in muscle size for individual RT-induced improvements in muscle strength of young men under normal physiological conditions (i.e., healthy and with adequate protein and energy intake; for details, see (34)).

Understanding the quantitative determinants of RT-induced strength gains is essential to refine RT and optimize the desired outcomes. The current study provides novel evidence that individual strength gains are much more strongly related to muscle growth ($r = 0.89$ – 0.92) than previously thought. Previous studies have found surprisingly low, insignificant or moderately significant relationships between the changes in muscle size and strength after RT using between-participant correlation ($r = 0.03$ – 0.53). These relatively low correlations may have been due to data collection limitations, including studies utilizing low-resolution muscle size measurements (e.g., ultrasound or DXA) (20,22), inducing limited muscle growth (e.g., $<7\%$) (20), or single measurements of strength at each time point. Consequently, to overcome these limitations, the current study involved a 15-wk (45 sessions) RT intervention that induced, on average, 12% muscle growth measured with high-resolution

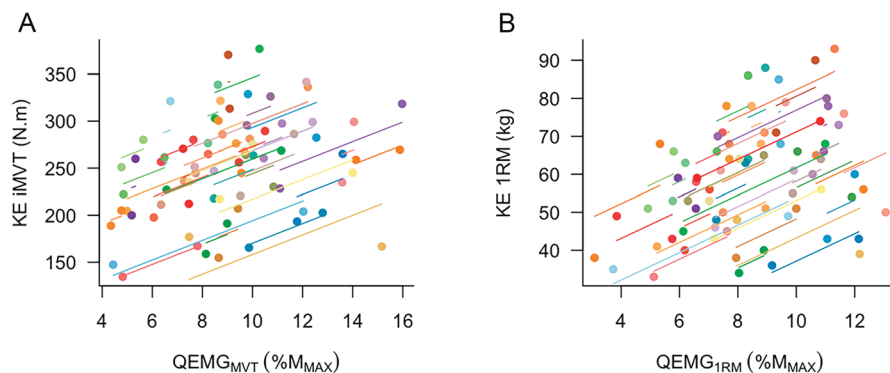


FIGURE 2—Within-participant relationship ($n = 39$) between changes in quadriceps (Q) neuromuscular activation measured with surface EMG during these tasks and the changes in (A) KE iMVT ($r = 0.58$) and (B) KE 1RM over 15 wk of RT ($r = 0.56$). All figures display observations from the same participant using the same color, with corresponding lines to show the repeated-measures correlation fit for each participant.

and highly reliable MRI (i.e., within-participant coefficient of variation <1%) (34) and duplicate measurements of strength both pre- and post-RT. Perhaps due to these methodological strengths, the between-participant correlation of the percentage changes in isometric strength and muscle size in the current study ($r = 0.60$) was the largest correlation that, to our knowledge, has so far been reported. Nonetheless, this is only slightly higher than our previous studies that also used MRI after 12 wk of RT and between participant correlation ($r = 0.46$ – 0.53) (9,24).

In contrast, repeated-measures correlation for both independent measures of muscle strength revealed very strong relationships with muscle growth (isometric strength, $r = 0.92$; 1RM lifting strength, $r = 0.89$) and thus much stronger relationships than previously documented. Therefore, the analytical approach appears imperative to explaining the different findings (22,32). Repeated-measures correlation (i.e., within-participant) is a superior approach to Pearson's between-participant correlation for handling repeated-measures data by accounting for the dependency between observations and utilizing multiple data points per participant (33). This approach assesses the overall or common intraindividual association between two measures, thus enhancing the sensitivity of the analysis. It provides the optimal linear fit for each participant using parallel regression lines (the same slope) with different intercepts (31). In addition, percentage change data from baseline have been found to have relatively low statistical power (37) and rely on the assumption that the absolute changes in a variable are related to the baseline value, which was not the case for strength or muscle size measures in the current study.

The purpose of this study was not to assess causality, rather to better delineate the relationship of muscle growth and the changes in neuromuscular activation with RT-induced strength gains. However, we recognize that there is considerable interest in the causal determinants of RT-induced strength gains, which raises the question of whether the very strong relationships between muscle growth and strength gains that we have observed are coincidental or causal. A range of observations, including the strong relationship between human muscle size and strength both pre- and post-RT (16,38), group-level changes in muscle size and strength gains that are broadly coupled with aging (39), immobilization (40), and testosterone administration (41), as well as the superior strength of highly trained individuals that appears to be due to their large muscle size (38), and now also rigorous experimental and statistical methods revealing a very strong relationship between individual muscle growth and strength gains, may suggest that this relationship may be primarily causal. However, causality does need to be demonstrated, even though this remains a significant research challenge as a targeted manipulation of muscle growth in isolation during RT (i.e., that can be demonstrated not to affect the other potential physiological contributors to strength gains) has not yet been identified.

Although a range of adaptations within the nervous system occur with regular RT (8), the contribution of these changes to motor performance has been opaque (14). Using between-

participant correlation, our previous work also found contradictory results of significant (24) and nonsignificant (9) relationships between the changes in neuromuscular activation and strength gains. The moderate within-participant correlations ($r = 0.56$ – 0.58) between the change in neuromuscular activation (assessed by sEMG) and strength in the current study in a larger population ($n = 39$) over a longer period of RT (15 wk) provide novel evidence that neural adaptations do contribute to strength changes.

Using linear mixed models, the current study also examined the combined influence of muscle size and activation changes as predictors of strength gains. We found both variables to contribute significantly to the increases in strength, but the changes in muscle size were the predominant contributing predictor for strength improvements in both strength tasks, being >5 times as strong as a predictor than neuromuscular activation. Our previous study, using a less robust approach (multiple linear regression on percentage changes), found neuromuscular activation to be the primary determinant, and muscle growth the secondary determinant, of strength gains (24). Besides the more robust statistical approach, the current study involved a longer RT period (15 vs 12 wk) that might have contributed to muscle growth being the primary determinant of strength gains in the current study. Although there is little robust evidence, neural changes are widely considered to be the primary explanation for the improvements in strength in the first weeks of a RT program (8,14), whereas morphological changes, including muscle size, may become progressively more important after the first few weeks of RT. Therefore, it is possible that the association of different adaptations with strength gains changes over time throughout a training program (i.e., with RT duration). However, the lack of longitudinal data at multiple time points (i.e., repeated mechanistic measurements) limits our understanding of the timecourse of the adaptive processes that occur during a RT program and, consequently, also their contribution to strength gains. Moreover, the current findings emphasize that to optimize strength gains with RT, it appears desirable to maximize increases in muscle size.

This research had some limitations. Increases in muscle strength arise from a myriad of factors beyond muscle growth and agonist activation, such as changes in muscle architecture, muscle composition (i.e., fiber type), contractile specific tension, antagonist and stabilizer muscle activation (8). The purpose of the current study was not to address the influence of all these factors simultaneously, but to examine the effect of two potentially key factors (muscle growth and neuromuscular activation) with rigorous experimental and statistical methods. Nonetheless, evaluating only two factors may be considered an oversimplified approach to a complex system. The measure of agonist neuromuscular activation in this study was sEMG amplitude, which has some well-documented limitations (16,42,43) and could potentially have contributed to the weaker association of changes in neuromuscular activation than muscle growth that we have observed. Although an array of other techniques have been used to assess neuroplasticity after RT (14) and could potentially contribute more to the

explained variance in strength gains than sEMG amplitude, the rationale for this remains unclear.

CONCLUSIONS

Using rigorous experimental and statistical techniques, particularly within-participant analysis, our results demonstrated that the changes in strength after 15 wk of RT (45 sessions) were very strongly related to muscle growth and moderately associated with neuromuscular activation. Furthermore, muscle size and neuromuscular activation both contributed to the prediction of strength changes after RT, with muscle size changes as the predominant contributing predictor. These findings suggest that muscle growth is the major adaptation relevant to the increases in strength of individual participants

after RT and supports a focus on hypertrophic stimuli as a component of RT.

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